

1 Experimental setup

Sequences are drawn from a mixture of three groups. Group g generates inputs $x \sim \mathcal{N}(0, \Lambda_g)$ with trace-normalised covariance, and all groups share the per-sequence task vector ω , so $y = \omega^\top x$ throughout. Groups therefore differ *only* in the shape of their input covariance, summarised by the participation ratio $\text{PR} = (\text{tr } \Lambda)^2 / \text{tr}(\Lambda^2)$. The flattest group, z_3 , is the forget set.

A linear-attention model performs in-context regression on these sequences. We train two shadow ensembles of 512 models each: a *full* model trained on all three groups, and a *retrain oracle* trained on z_1, z_2 only. At evaluation the z_3 tokens in a frozen probe are corrupted — a prompt-time edit, with no weight update — under three families:

$$\begin{aligned} \text{C1 (label noise)} : (x_f, y_f) &\rightarrow (x_f, y_f + \epsilon), & \epsilon &\sim \mathcal{N}(0, \sigma^2) \\ \text{C2 (input noise)} : (x_f, y_f) &\rightarrow (x_f + \epsilon, y_f), & \epsilon &\sim \mathcal{N}(0, \sigma^2 I_D) \\ \text{C3 (both)} : (x_f, y_f) &\rightarrow (x_f + \epsilon_1, y_f + \epsilon_2), & \epsilon_1 &\perp \epsilon_2 \end{aligned}$$

The adversary observes the sign-aligned residual $\text{sign}(y)(\hat{y} - y)$ at a fixed probe point and ranks the two hypotheses across the shadow axis; we report the Mann–Whitney AUC averaged over the probe. Both hypotheses are scored on the *same* corrupted prompt with the same noise draw, so that edit-detection cannot be mistaken for membership evidence. Preservation is measured as $\varepsilon = \text{KL}(p_{\text{oracle}} \| p_{\text{unlearned}})$ on the forget-population residual law.

Data		
input dimension	D	4
context length	N	31
groups		z_1, z_2, z_3
forget group		z_3
eigenbasis		identity (shared across groups)
Λ_{z_1} eigenvalues	PR = 1.905	0.70, 0.15, 0.10, 0.05
Λ_{z_2} eigenvalues	PR = 2.381	0.60, 0.20, 0.10, 0.10
Λ_{z_3} eigenvalues	PR = 3.333	0.40, 0.30, 0.20, 0.10
data seed		0
Probe		
probe points	P	64
token counts		$z_1:10, z_2:10, z_3:11$
probe seeds		777, 778, 779
Training		
architectures		ATTN-S, ATTN-M
shadow models per hypothesis	S	512
batch per shadow		8
steps		6000
optimiser		SGD, momentum 0.9
learning rate		0.005
gradient clipping		5.0, per shadow
initialisation scale		0.05
training seeds		1, 2, 3
Evaluation		
seed combinations		$3 \times 3 = 9$
bootstrap resamples		200
shared-noise repeats		16
C1, C2 grid		21 points, $\sigma^2 \in [0, 100]$, log-spaced
C3 grid		9 points, $\sigma^2 \in [0, 10]$

Table 1: Complete experimental settings.

2 Results

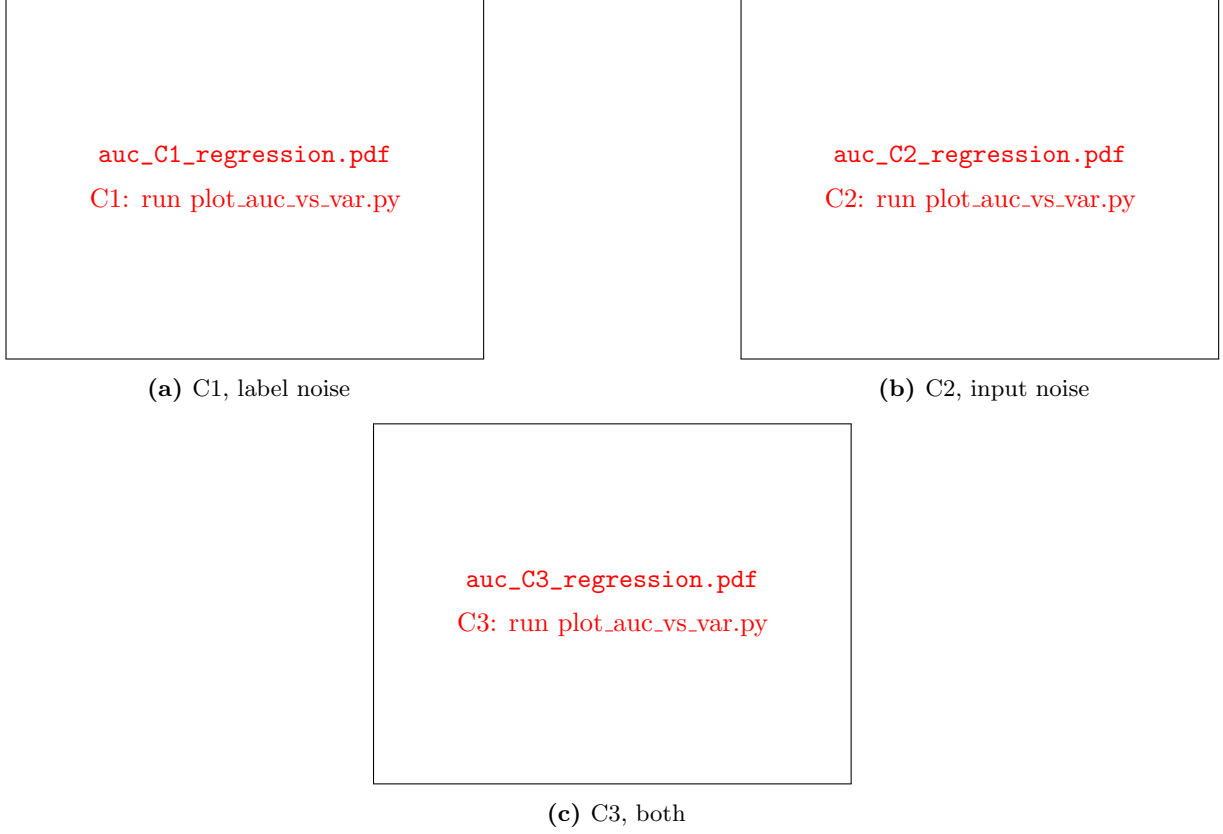
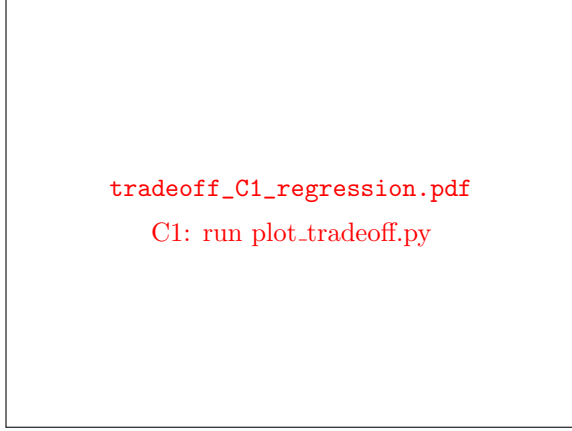


Figure 1: Membership AUC against corruption variance σ^2 . Bands give the min-max range over nine (training seed, probe seed) combinations. Dotted line: chance. Thin solid line: closed-form prediction. Dotted coloured line: shared-noise control.

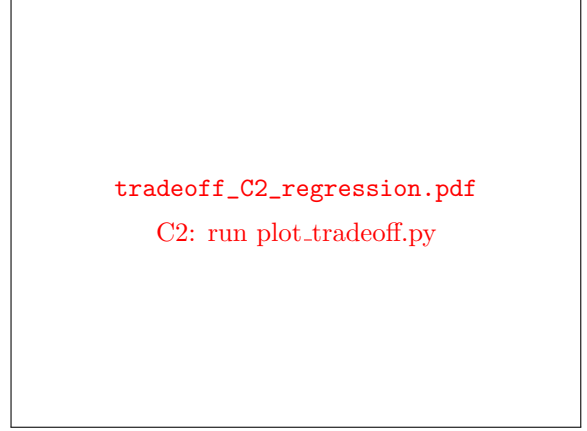
Figure 1 shows AUC against σ^2 for each corruption family. At $\sigma^2 = 0$ the adversary achieves $\text{AUC} = \mathbf{X.XXX}$, and AUC rises monotonically toward chance as corruption strength increases, reaching $\mathbf{X.XXX}$ at $\sigma^2 = 100$. Measurements agree with a closed-form Gaussian prediction derived from the read-out algebra to within $\mathbf{X} \times 10^{-4}$ across the grid.

Two observations qualify this. First, a shared-noise control — a single corruption draw broadcast across the ensemble, holding variance inflation fixed — attributes $\mathbf{XX}\%$ of the movement toward chance to *masking* rather than removal at $\sigma^2 \leq 1$, falling to $\mathbf{XX}\%$ at $\sigma^2 = 100$. Below unit variance the adversary is defeated by inflated within-ensemble spread, not by removal of forget-set information.

Second, and shown in Figure 2, ε is minimised at $\sigma^2 \approx \mathbf{X}$, where $\varepsilon = \mathbf{X.XXe-X}$ and $\text{AUC} = \mathbf{X.XXX}$. At the corruption strength bringing AUC closest to chance, ε has risen to $\mathbf{X.XX}$ — a factor of $\mathbf{X.X} \times 10^3$. Within four decades of corruption strength there is no operating point at which the adversary is defeated and the model remains close to the retrain oracle.



(a) C1



(b) C2

Figure 2: Removal versus preservation. AUC (solid, left axis) and ε (dashed, right axis, log scale). Vertical markers: the ε minimum and the point of closest approach to chance.

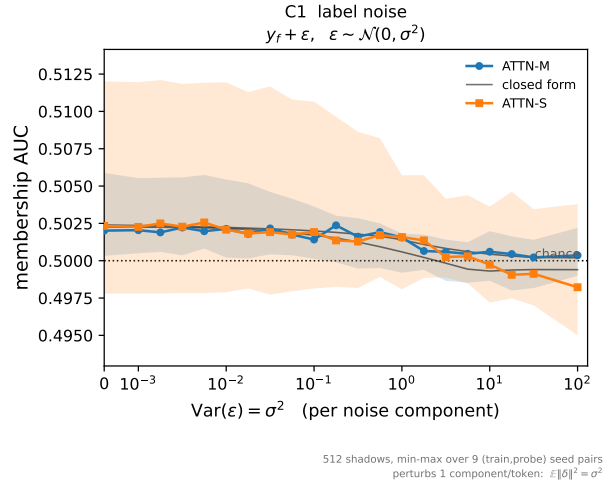


Figure 3: Null control. All three groups are given the same covariance spectrum and the same eigenbasis, making them one distribution; the full and oracle models therefore train on indistinguishable data and no membership signal exists. AUC sits at chance across the entire grid (**0.5XX** and **0.5XX** for the two architectures).